

EyeMVP: OCT-Informed Fundus Representation Learning via Paired CFP–OCT Pretraining

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Abstract—Color fundus photography (CFP) is the mainstay for large-scale retinal screening, yet its diagnostic capacity is constrained by the lack of depth-resolved structural information. Optical coherence tomography (OCT) provides cross-sectional retinal anatomy, but is less accessible in population-level screening. Here, we present EyeMVP, a cross-modal retinal foundation model that uses paired CFP–OCT pretraining to learn OCT-informed CFP representations. EyeMVP is pretrained on 674,893 strict same-eye same-day paired CFP–OCT image triples from 112,642 patients across eight hospitals in China. The model uses cross-modal masked reconstruction to enrich CFP representations with OCT-associated supervision, while requiring only CFP images at inference. To accommodate the non-aligned imaging geometry between en-face CFP and cross-sectional OCT, EyeMVP combines source-constrained cross-attention with CFP-derived structural masks. Across 15 downstream tasks, including classification, segmentation, and few-shot adaptation, EyeMVP performs on par with or better than representative retinal foundation models overall, with consistent gains on tasks involving macular and optic nerve structure. For CFP-challenging macular diseases, EyeMVP achieves an AU-ROC of 0.923 for macular edema and 0.867 for myopic mac-

ular schisis. Qualitative t-SNE visualization of the frozen encoder embeddings further shows that EyeMVP produces more class-coherent CFP representations than competing foundation models. In an exploratory reader study, EyeMVP exceeds junior and intermediate ophthalmologist groups but does not reach senior ophthalmologist performance on macular edema, while showing numerically higher balanced accuracy than all reader groups on myopic macular schisis. These results suggest that pixel-level cross-modal reconstruction can enrich CFP representations with OCT-associated supervision, providing a practical route toward stronger CFP-based retinal analysis in screening settings.

Index Terms—Foundation model, cross-modal learning, optical coherence tomography, Color fundus photography.

I. INTRODUCTION

COLOR fundus photography (CFP) is the mainstay of large-scale retinal screening because it is inexpensive, rapid, and widely available. However, CFP is a two-dimensional en-face projection of the retinal surface and therefore provides only indirect information about depth-resolved retinal structure. Many clinically important abnormalities, including macular edema, myopic macular schisis, intraretinal fluid, and optic nerve head deformation, are defined or confirmed by cross-sectional anatomy rather than by surface appearance alone. Optical coherence tomography (OCT) provides this layer-resolved structural information and has become central to modern retinal diagnosis, but its availability in population-level screening and resource-limited settings remains more restricted than CFP [1], [2]. This gap motivates a practical representation learning question: can paired CFP–OCT data be used during pretraining to enrich CFP representations with OCT-associated supervision, while requiring only CFP images at inference?

Recent retinal foundation models have shown that self-supervised pretraining can substantially improve the transferability of ophthalmic image representations [3]–[5]. Unimodal masked autoencoders such as RETFound [6] learn strong CFP features from large-scale fundus collections and generalize across multiple downstream tasks. Nevertheless, models trained only on CFP are limited by the information available in the input modality. They can learn rich appearance and texture patterns, but they receive no direct learning signal from cross-sectional retinal anatomy. This is particularly limiting for CFP–

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challenging diseases whose visual signs are subtle, indirect, or partially occult in fundus photographs.

A natural strategy is to incorporate OCT during pretraining. Existing multimodal ophthalmic models, including VisionFM [7], EyeCLIP [8], Eyefound [9], and MIRAGE [10], have explored multimodal representation learning across retinal imaging modalities. Many of these approaches rely on image-level contrastive alignment or related global objectives, following the broader success of contrastive image-language pretraining [11]. Such objectives are effective for learning semantic correspondence between modalities, but they do not explicitly exploit the pixel- or patch-level structural information available in paired CFP–OCT data. For diseases whose diagnostic evidence depends on localized macular or optic nerve structure, global alignment may underuse the most informative cross-modal supervision.

Masked reconstruction provides a more direct way to use cross-modal supervision between modalities. In principle, reconstructing OCT from CFP tokens can encourage the model to encode CFP features that are predictive of underlying retinal structure. However, applying standard multimodal masked autoencoding to CFP–OCT pairs is not straightforward. Frameworks such as MAE [12] and MultiMAE [13] assume that modalities share a common spatial grid, as in RGB-depth or RGB-segmentation pairs. CFP and OCT violate this assumption: CFP represents the retinal surface in an en-face coordinate system, whereas OCT samples cross-sectional depth structure. A CFP patch and an OCT patch with the same image index do not correspond to the same physical retinal location. This non-aligned imaging geometry makes naive patch-level cross-modal reconstruction prone to learning superficial correlations rather than useful structural correspondence.

In this work, we present EyeMVP, a cross-modal retinal foundation model designed to learn OCT-informed CFP representations from strict same-eye same-day paired CFP–OCT images. EyeMVP uses cross-modal masked reconstruction as the pretraining signal, but adapts it to the non-aligned geometry of CFP and OCT through two design choices. First, a source-constrained cross-attention decoder restricts cross-modal reconstruction to source-modality features, creating an information bottleneck that discourages local texture shortcuts. Second, CFP-derived structural masks provide anatomical anchors, including the optic disc, macular region, and major vessels, to ground cross-modal learning without requiring explicit patch-level CFP–OCT registration.

We pretrain EyeMVP on 674,893 strict same-eye same-day paired CFP–OCT image triples from 112,642 patients across eight hospitals in China. The pretrained CFP encoder is evaluated on 15 downstream tasks, including retinal disease classification, dense segmentation, and few-shot adaptation. Across these tasks, EyeMVP is competitive with or improves over representative retinal foundation models, with the largest gains on macular and optic nerve tasks where structural information is clinically important. For CFP-challenging macular disease classification, EyeMVP achieves an AUROC of 0.923 for macular edema and 0.867 for myopic macular schisis. An exploratory reader study further suggests that the learned CFP representation may provide useful support for macular

conditions that are difficult to assess from fundus photographs alone.

The main contributions of this work are:

- We propose EyeMVP, an OCT-informed CFP representation learning framework that uses paired CFP–OCT data during pretraining while requiring only CFP images at downstream inference.
- We introduce a cross-modal masked reconstruction strategy tailored to the non-aligned imaging geometry of CFP and OCT, using source-constrained cross-attention to encourage prediction from the source modality rather than target-modality shortcuts.
- We incorporate CFP-derived anatomical masks as structural layout guidance during pretraining, providing coarse fundus landmarks without requiring explicit patch-level CFP–OCT registration.
- We pretrain EyeMVP on 674,893 strict same-eye same-day CFP–OCT image triples from 112,642 patients and evaluate it across 15 downstream settings, showing consistent gains over representative retinal foundation models, especially on macular and optic nerve tasks.

II. RELATED WORK

A. OCT-Enhanced Fundus Image Analysis

CFP and OCT provide complementary views of retinal disease. CFP is widely available and captures en-face retinal appearance, whereas OCT provides cross-sectional structural information that is often required to confirm or quantify macular and optic nerve abnormalities [2], [14]. A line of prior work has therefore investigated whether OCT-derived labels or measurements can improve fundus-based analysis. Varadarajan et al. [15] trained deep learning models to predict OCT-derived center-involved diabetic macular edema and retinal fluid from fundus photographs, showing that CFP can contain indirect visual cues associated with OCT-defined disease. Similarly, Medeiros et al. [16] used OCT retinal nerve fiber layer measurements to supervise a model that predicts glaucomatous structural damage from optic disc photographs. These studies established the feasibility of learning OCT-associated information from CFP, but they were primarily task-specific supervised models rather than general-purpose pretraining frameworks.

More recent work has explored cross-modal knowledge transfer between fundus and OCT images. MultiEYE and OCT-CoDA [17] formulate an OCT-enhanced disease recognition setting in which OCT information is used during training to improve fundus-only inference, including through concept-based distillation from OCT to CFP. This direction is closely aligned with the practical motivation of our work: OCT should enrich fundus representations during model development, but downstream deployment should remain compatible with CFP-only screening. EyeMVP differs in two respects. First, it learns from strict same-eye same-day paired CFP–OCT images through pixel-level cross-modal masked reconstruction rather than task-specific disease distillation. Second, it aims to produce a reusable CFP encoder for classification, segmentation, and few-shot adaptation, rather than a model specialized to one disease recognition benchmark.

B. Retinal Foundation Models

Foundation models have recently become an important direction in ophthalmic image analysis. RETFound [6] demonstrated that masked autoencoding on large-scale retinal images can produce transferable representations for disease detection and segmentation. VisionFM [7] further scaled ophthalmic pretraining across multiple image modalities, tasks, disease categories, devices, and demographic groups, supporting a generalist ophthalmic AI setting. Vision-language models such as FLAIR [18] and EyeCLIP [8] incorporate textual supervision or multimodal contrastive learning to improve zero-shot, few-shot, and long-tail disease performance. Eyefound [9] and MIRAGE [10] also expand foundation modeling toward broader ophthalmic or retinal modality coverage.

Despite this progress, most existing retinal foundation models are optimized for either unimodal representation learning or global multimodal alignment. Unimodal CFP models can learn strong appearance features but do not receive direct supervision from cross-sectional retinal structure. Multimodal contrastive or vision-language models improve semantic alignment across images and text, but their objectives usually operate at the image or report level and therefore do not explicitly exploit the local structural correspondence available in paired CFP–OCT data. EyeMVP is complementary to these efforts: it focuses on using paired OCT as a pretraining signal to enrich CFP representations with structural information while retaining CFP-only inference.

C. Multimodal Masked Reconstruction Pretraining

Masked autoencoders have provided a simple and scalable framework for self-supervised visual representation learning [3]. MAE [12] trains a vision transformer to reconstruct randomly masked image patches and has been widely adopted in medical imaging because reconstruction encourages the model to capture fine-grained spatial structure. MultiMAE [13] extends this idea to multimodal and multitask settings by reconstructing multiple outputs, such as RGB, depth, and semantic segmentation, from partially visible modality tokens. Related cross-modal masked reconstruction methods have also been explored in broader multimodal pretraining settings [19], and in medical vision-language pretraining; for example, M³AE [20] reconstructs masked image and text tokens to learn cross-modal medical representations.

However, existing multimodal masked autoencoding frameworks are most natural when modalities share a common spatial coordinate system or can be treated as aligned token sequences. This assumption holds for many RGB-depth-segmentation settings, but it is violated by CFP–OCT pairs. CFP is an en-face projection of the retinal surface, whereas OCT samples cross-sectional depth structure; therefore, patch indices across the two modalities do not denote the same physical retinal location. Naively applying multimodal masked reconstruction to CFP–OCT can encourage shortcut correlations or unstable cross-modal optimization rather than robust structural representation learning. EyeMVP adapts masked reconstruction to this asymmetric imaging geometry through

source-constrained cross-attention and CFP-derived structural anchors.

III. METHODS

A. Pretraining Dataset Construction

Raw multimodal retinal imaging data were collected from eight tertiary hospitals across China between August 2015 and December 2025. The archive contained 3,058,942 images from 112,642 participants (59,187 females and 53,455 males; mean age 57.5 years, SD 14.6, range 1–95; all of Chinese ethnicity). CFP images were acquired using five platforms, including Zeiss CLARUS 500, Canon CR-2, Topcon TRC, SYSEYE, and Airdoc systems. OCT volumes were acquired using Zeiss CIRRUS, Heidelberg Spectralis, and SVision systems. Site-level ethics approval was coordinated by the lead center, Beijing Tongren Hospital, and all data were fully de-identified before analysis.

CFP examinations included both optic-disc-centered and macula-centered fundus photographs. OCT examinations included both optic nerve head and macular scan protocols, reflecting routine clinical imaging for glaucoma and macular disease assessment. CFP and OCT examinations were linked using patient identifier, eye laterality, and acquisition date. A valid pair required CFP and OCT to be acquired from the same eye on the same calendar day and to pass metadata and image-quality checks. We refer to this criterion as strict same-eye same-day pairing. Eye laterality was determined from DICOM metadata and checked against fundus image orientation. For each retained patient-eye visit, representative OCT B-scans were sampled from the corresponding OCT volume and linked to the CFP image from the same visit. CFP-derived structural masks were used to provide coarse fundus-layout information during pretraining.

A four-stage quality control pipeline was applied. First, each raw image was screened using an automated image quality assessment algorithm [21]; images with severe blur, motion artifacts, under- or over-exposure, pupil occlusion, incomplete retinal field, or poor OCT signal were removed. Second, same-day same-eye CFP–OCT examinations were identified from DICOM timestamps and laterality tags. Third, candidate pairs with missing or inconsistent metadata, implausible acquisition timestamps, insufficient retinal coverage, or laterality mismatch were excluded. Fourth, patients included in private downstream evaluation datasets were excluded from the pretraining pool to reduce data leakage risk.

For each eligible patient-eye visit, CFP images were paired with representative B-scans from the corresponding OCT volume. Rather than using only a single central B-scan, we sampled 3–4 representative B-scans from each retained OCT volume to provide limited spatial coverage of the acquired scan region. This produced 189,801 eligible patient-eyes and 674,893 final CFP–OCT image triples, with a mean of 3.56 OCT frames per patient-eye. Each triple consisted of a CFP image, a paired OCT B-scan, and a CFP-derived structural mask. A random 5% subset of the constructed triples was manually reviewed by clinicians to verify same-eye pairing, image quality, and segmentation plausibility.

All CFP and OCT images were resized to 512×512 . OCT B-scans were represented as three-channel grayscale images to use the same reconstruction interface as CFP. CFP-derived structural masks were resized to 256×256 and encoded as discrete anatomical labels. A site-level breakdown of the pretraining data and imaging equipment is provided in Table I.

B. CFP-Derived Structural Masks

CFP-derived structural masks (CFP-Seg) were used as an auxiliary input during pretraining. Each mask contains four classes: background, optic disc, foveal region, and major retinal vessels. The masks were generated offline from CFP images using task-specific segmentation networks, including U-Net [22] for optic disc and foveal region segmentation, and UNet++ or Attention U-Net [23]–[25] for retinal vessel segmentation. These networks were trained on public fundus segmentation datasets [26] and fine-tuned on held-out in-house annotations that were not included in pretraining or downstream evaluation. The segmentation outputs were binarized and merged into a single four-class mask for each CFP image.

CFP-Seg does not provide disease labels. Its role is to provide coarse anatomical layout information from the CFP image itself. Because CFP and OCT are acquired in different imaging geometries, CFP-Seg supplies explicit fundus landmarks that help condition cross-modal reconstruction on the observed CFP field.

In this setting, CFP-Seg helps identify the optic disc, foveal region, and major vessel trajectories in the CFP image, providing structural context about the current fundus field and spatial layout. CFP-Seg is used only during pretraining and is discarded during downstream inference. Its contribution is evaluated by removing this auxiliary input in ablation experiments.

C. EyeMVP Architecture

EyeMVP uses three input modalities during pretraining: CFP, OCT, and CFP-Seg. Each modality is divided into non-overlapping patches and projected into token embeddings by a modality-specific patch embedding layer. CFP and OCT inputs use patch size 32×32 , and CFP-Seg uses patch size 16×16 . The token sequences are then processed by a shared ViT-L encoder [27]. Thus, different modalities have separate input projections but share the same encoder parameters.

The framework contains five lightweight decoders. Three are intra-modal reconstruction decoders for CFP, OCT, and CFP-Seg:

$$C \rightarrow C, \quad O \rightarrow O, \quad S \rightarrow S.$$

These decoders reconstruct masked patches of their own modality from encoded token features and are used throughout pretraining.

The remaining two decoders are cross-modal reconstruction decoders:

$$C \rightarrow O, \quad O \rightarrow C.$$

They are introduced only in the second training stage. The CFP-to-OCT decoder uses CFP encoded tokens to reconstruct OCT patches, and the OCT-to-CFP decoder uses OCT encoded

tokens to reconstruct CFP patches. These decoders provide an explicit cross-modal prediction objective, which is not guaranteed by intra-modal reconstruction alone.

D. Dirichlet Masking Strategy

During pretraining, we use a Dirichlet masking strategy to sample visible token allocation across modalities. For each training sample, modality-wise visible token proportions are drawn from a Dirichlet distribution:

$$(r_C, r_O, r_S) \sim \text{Dirichlet}(\alpha_C, \alpha_O, \alpha_S), \quad (1)$$

where r_C , r_O , and r_S denote the visible token proportions for CFP, OCT, and CFP-Seg. Given the total visible token budget, visible tokens are sampled within each modality according to these proportions, and the remaining tokens are masked.

This strategy exposes the shared encoder to variable modality availability. Compared with using a fixed masking ratio for each modality, Dirichlet masking provides a broader range of reconstruction conditions and reduces dependence on a single fixed modality configuration. During Stage II, the sampling is constrained to retain sufficient visible tokens from the source modality so that CFP-to-OCT and OCT-to-CFP reconstruction receive stable source information.

E. Source-Constrained Cross-Modal Decoding

Intra-modal reconstruction alone does not guarantee that the shared encoder learns useful CFP–OCT correspondence. When CFP is reconstructed from CFP features and OCT from OCT features, the model can solve the pretext task largely through within-modality cues. For high-capacity encoders and lightweight decoders, local texture and modality-specific appearance statistics can often support good reconstruction. Thus, a shared encoder with multiple intra-modal heads may encourage feature sharing, but it does not explicitly require CFP features to predict OCT structure.

To encourage cross-modal interaction, EyeMVP introduces CFP-to-OCT and OCT-to-CFP reconstruction in Stage II. A naive cross-modal decoder may still exploit shortcuts if target-modality visible tokens or mixed multimodal tokens are available during decoding. In the CFP–OCT setting, such shortcuts are undesirable because the modalities are not patch-wise registered; local texture correlations may reduce reconstruction loss without learning useful structural correspondence.

We therefore use source-constrained cross-attention. For CFP-to-OCT reconstruction, OCT positional tokens are used as Queries, while CFP encoded tokens are used as Keys and Values:

$$\text{Attn}_{C \rightarrow O} = \text{softmax} \left(\frac{Q_O K_C^\top}{\sqrt{d}} \right) V_C. \quad (2)$$

For OCT-to-CFP reconstruction, CFP positional tokens are used as Queries, while OCT encoded tokens are used as Keys and Values:

$$\text{Attn}_{O \rightarrow C} = \text{softmax} \left(\frac{Q_C K_O^\top}{\sqrt{d}} \right) V_O. \quad (3)$$

In both directions, the target modality provides only positional query tokens and does not provide target visible tokens

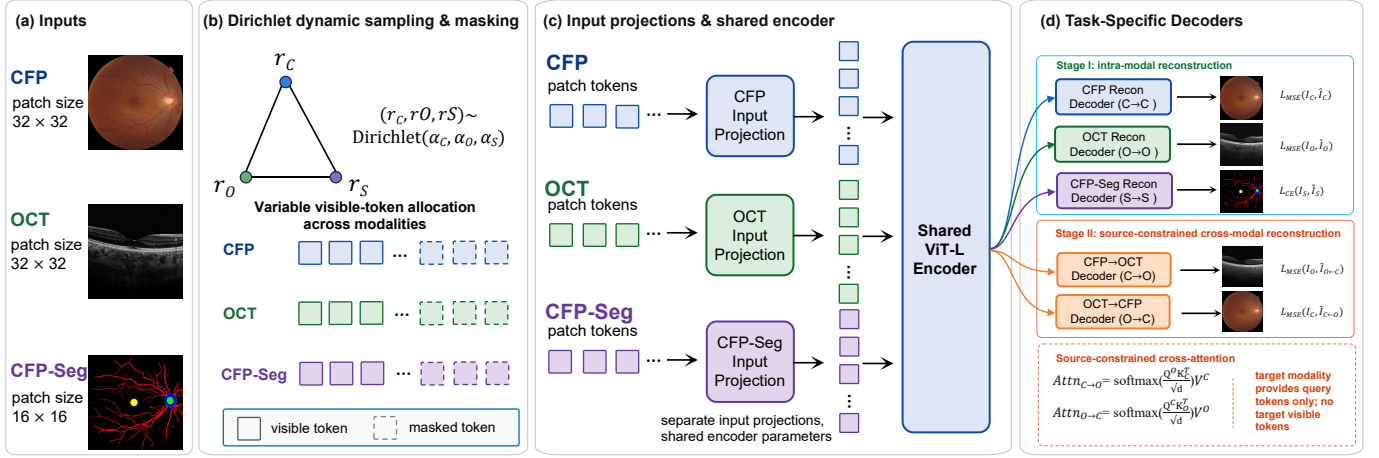


Fig. 1. Overview of the EyeMVP pretraining framework. Each pretraining sample contains a CFP image, a paired OCT B-scan, and a CFP-derived structural mask (CFP-Seg). Dirichlet sampling dynamically assigns visible token ratios across modalities, and modality-specific input projections map the visible tokens into a shared ViT encoder. Stage I uses three intra-modal decoders to reconstruct CFP, OCT, and CFP-Seg targets. Stage II introduces two source-constrained cross-modal decoders for CFP-to-OCT and OCT-to-CFP reconstruction. CFP and OCT reconstruction losses use mean-squared error, whereas CFP-Seg reconstruction uses cross-entropy.

TABLE I

PER-CENTER BREAKDOWN OF THE PRETRAINING DATASET. COHORT DEMOGRAPHICS AND DISEASE DISTRIBUTION ARE GIVEN IN SUPPLEMENTARY TABLES S1 AND S2.

Site	Patients	CFP	OCT
Beijing Tongren Hospital	57,830	CLARUS/CR-2/SYSEYE/Airdoc	Spectralis/SVision
Fangshan District Hospital	26,239	CR-2	Spectralis/SVision
Dalian No.3 Hospital	9,826	CR-2	Spectralis
Baoji Central Hospital	5,837	CR-2	Spectralis
Wuxi No.2 Hospital	5,408	TRC/CR-2	CIRRUS/Spectralis
Shenzhen Eye Hospital	3,647	CR-2/TRC	Spectralis
Second XiangYa Hospital	2,653	TRC/CR-2	CIRRUS/Spectralis
Beijing Friendship Hospital	1,202	CR-2	Spectralis
Total	112,642	5 platforms	3 platforms

to the cross-modal decoder. Therefore, the decoder must use source-modality features to predict the target modality. This design encourages cross-modal prediction while reducing the tendency to rely on target-modality shortcuts.

F. Two-Stage Pretraining

We adopt a two-stage schedule to stabilize optimization. In Stage I, from epoch 0 to 400, only the three intra-modal reconstruction objectives are optimized. CFP, OCT, and CFP-Seg tokens are randomly masked, encoded by the shared encoder, and reconstructed by their corresponding intra-modal decoders. This stage provides a basic initialization for the shared encoder and allows the model to learn CFP appearance, OCT structure, and CFP-derived anatomical layout.

Stage I is not intended to enforce CFP-OCT alignment. Since each modality is reconstructed from its own encoded features, the model can solve the task largely using within-modality information. Thus, Stage I learns useful basic representations, but does not explicitly require CFP features to predict OCT structure or OCT features to predict CFP appearance.

In Stage II, from epoch 401 to 800, the two source-constrained cross-modal decoders are introduced. The model

continues to optimize the three intra-modal reconstruction losses and additionally optimizes CFP-to-OCT and OCT-to-CFP reconstruction losses. This stage preserves within-modality reconstruction while adding an explicit cross-modal prediction objective.

1) Training Objectives: The Stage I objective is a weighted sum of per-modality masked reconstruction losses:

$$\mathcal{L}_{\text{intra}} = \lambda_C \mathcal{L}_{\text{MSE}}(I_C, \hat{I}_C) + \lambda_O \mathcal{L}_{\text{MSE}}(I_O, \hat{I}_O) + \lambda_S \mathcal{L}_{\text{CE}}(I_S, \hat{I}_S), \quad (4)$$

where I_C , I_O , I_S denote CFP, OCT, and CFP-Seg patches; $\hat{\cdot}$ denotes reconstructed counterparts; and $\lambda_C, \lambda_O, \lambda_S$ balance modality contributions. Losses are computed over the masked token index set $\mathcal{M}$ ($N=|\mathcal{M}|$):

$$\mathcal{L}_{\text{MSE}}(x, \hat{x}) = \frac{1}{N} \sum_{i \in \mathcal{M}} \|x_i - \hat{x}_i\|_2^2, \quad (5)$$

$$\mathcal{L}_{\text{CE}}(y, \hat{y}) = -\frac{1}{N} \sum_{i \in \mathcal{M}} \sum_{c=1}^C y_{i,c} \log \hat{y}_{i,c}. \quad (6)$$

The Stage II cross-modal objective reconstructs each modal-

ity conditioned on the encoded features of the other:

$$\mathcal{L}_{\text{cross}} = \lambda_{C \rightarrow O} \mathcal{L}_{\text{MSE}}(I_O, \hat{I}_{O \leftarrow C}) + \lambda_{O \rightarrow C} \mathcal{L}_{\text{MSE}}(I_C, \hat{I}_{C \leftarrow O}), \quad (7)$$

where $\hat{I}_{O \leftarrow C}$ denotes OCT patches reconstructed from CFP features via the source-constrained decoder. The overall training objective is:

$$\mathcal{L} = \lambda_{\text{intra}} \mathcal{L}_{\text{intra}} + \lambda_{\text{cross}} \mathcal{L}_{\text{cross}}. \quad (8)$$

G. Downstream Adaptation

After pretraining, only the CFP branch is used for downstream evaluation. The OCT input, CFP-Seg input, and all reconstruction decoders are discarded. For classification, the pretrained CFP encoder is frozen and a linear classifier is trained. For dense segmentation, a lightweight decoder is attached to the frozen CFP encoder. For few-shot evaluation, the same protocol is repeated with limited labeled examples per class. For feature-space visualization, CFP embeddings are extracted from the frozen encoder and projected to two dimensions using t-SNE for qualitative inspection of representation quality.

IV. EXPERIMENTS AND RESULTS

A. Implementation Details

The shared encoder is a ViT-L model with embedding dimension $d=1024$, 24 transformer layers, and 16 attention heads. It is initialized from ImageNet MAE weights [12], [28]. Pretraining uses AdamW [29] with $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.05, cosine learning rate decay from 10^{-4} to 10^{-6} , and a 40-epoch linear warmup. The batch size is 256. Data augmentation includes random horizontal flipping and random rotation.

Loss weights are set to $\lambda_C=\lambda_O=1.0$ and $\lambda_S=0.5$ for intra-modal reconstruction. In Stage II, $\lambda_{C \rightarrow O}=\lambda_{O \rightarrow C}=0.5$ is used for the two cross-modal reconstruction losses. Pretraining is performed for 800 epochs, with Stage I occupying the first 400 epochs and Stage II the remaining 400 epochs.

B. Downstream Evaluation Datasets

The downstream benchmark comprises 15 dataset-level evaluation settings across two categories: CFP-based classification and dense segmentation. All downstream evaluation cases were excluded from the pretraining pool.

For CFP-based classification, we evaluated EyeMVP on 11 dataset-level settings. These included common retinal and optic nerve disease classification tasks: Messidor-2 for diabetic retinopathy [30], IDRID for proliferative diabetic retinopathy [31], IChallenge-AMD for age-related macular degeneration [32], ORIGA-650 and REFUGE for glaucoma [33], [34], PALM-2 for pathological myopia [35], ODIR5K for multi-disease recognition [36], and a private retinal vein occlusion cohort. We also evaluated three macular-disease cohorts that are difficult to assess from CFP alone: private OCT-confirmed datasets for macular edema, myopic macular schisis, and central serous retinopathy.

TABLE II

SUMMARY OF DOWNSTREAM EVALUATION DATASETS. THE BENCHMARK INCLUDES 11 CLASSIFICATION SETTINGS AND FOUR SEGMENTATION SETTINGS. REPRESENTATIVE IMAGE EXAMPLES ARE SHOWN IN SUPPLEMENTARY FIGS. S2 AND S3.

Task	Dataset	Source	Samples
DR cls	Messidor-2 [30]	Public	1,748
PDR cls	IDRID [31]	Public	516
AMD cls	IChallenge-AMD [32]	Public	400
Glaucoma cls	ORIGA-650 [33]	Public	650
Glaucoma cls	REFUGE [34]	Public	800
PM cls	PALM-2 [35]	Public	800
RVO cls	Private	Private	1,067
CSR cls	Private	Private	590
Multi-disease cls	ODIR5K [36]	Public	10,000
ME cls	Private	Private	600
MS cls	Private	Private	300
Disc/cup seg	REFUGE [34]	Public	1,200
Vessel seg	DRIVE [37]	Public	40
HE seg	IDRID [31]	Public	80
EX seg	e.opthaEX [38]	Public	48

DR: diabetic retinopathy; PDR: proliferative diabetic retinopathy; AMD: age-related macular degeneration; PM: pathological myopia; RVO: retinal vein occlusion; CSR: central serous retinopathy; ME: macular edema; MS: myopic macular schisis; HE: hemorrhage; EX: hard exudate.

For dense segmentation, we used four CFP-based datasets: REFUGE for optic disc/cup segmentation [34], DRIVE for retinal vessel segmentation [37], IDRID for hemorrhage segmentation [31], and e.opthaEX for hard exudate segmentation [38].

All private classification labels were assigned by two independent senior ophthalmologists, using OCT as the reference standard when applicable. Disagreements were resolved by a third adjudicating ophthalmologist. A concise summary of the downstream evaluation datasets is provided in Table II; detailed device information, image resolutions, and label distributions are provided in Supplementary Table S6.

C. Evaluation Protocols

After pretraining, only the CFP branch was used for downstream evaluation. For the private downstream classification datasets, each patient contributed only one study eye, and patients included in these datasets were excluded from the pretraining pool. For full-data classification, the pretrained CFP encoder was frozen and a linear classifier was trained on the extracted CFP features. Results were evaluated using five-fold cross-validation and reported as mean $\pm$ SD across folds. AUROC, balanced accuracy (BAcc), and average precision (AP) were reported. For few-shot evaluation, $K \in \{4, 8, 16\}$ labeled samples per class were sampled, and results were averaged over five random seeds.

For dense segmentation, the pretrained CFP encoder was frozen and a ConvNeXt-based decoder [39] was trained to predict pixel-wise masks. Full-data segmentation results were also evaluated using five-fold cross-validation and reported as mean $\pm$ SD across folds. Dice coefficient and intersection-over-union (IoU) were reported. For feature-space visualization, CFP embeddings were extracted from each frozen pretrained encoder and projected to two dimensions using t-SNE [40] with identical hyperparameters across all models; class labels

were used only to color the projected points and were not used during projection.

D. Classification Performance

Table III summarizes the full-data and few-shot classification AUROC across 11 dataset-level evaluation settings, covering common retinal diseases, optic nerve disease, and macular conditions that are difficult to identify from CFP alone. Under full-data linear probing, EyeMVP attains the highest or comparable AUROC on most datasets relative to RETFound [6], VisionFM [7], and EyeCLIP [8], with the clearest gains on structurally demanding macular and optic-nerve conditions. Against the strongest baseline, the largest improvements are on ORIGA-650 glaucoma (78.06% vs. 72.26%) and myopic macular schisis (86.67% vs. 82.00%), followed by AMD (94.81% vs. 92.01%) and macular edema (92.26% vs. 90.89%); on near-saturated PALM pathological myopia it is on par with the best CFP baselines (99.86% vs. 99.92%).

A consistent pattern holds in the few-shot setting ($K=4, 8, 16$): EyeMVP’s largest gains again fall on the macular and optic-nerve tasks where CFP alone is least informative, while it stays competitive elsewhere. At $K=16$ it reaches 81.71% AUROC on macular edema and 83.88% on myopic macular schisis (6.69 and 4.86 points above the strongest baseline) and 87.50% on REFUGE glaucoma, whereas on near-saturated pathological myopia the strong CFP baselines stay marginally ahead at low shot counts. This indicates that the cross-modal pretraining improves data efficiency, with benefits concentrated on structurally demanding conditions.

E. Dense Segmentation Performance

Table IV summarizes the full-data and few-shot segmentation results on four CFP-based dense prediction tasks, including optic disc/cup, retinal vessel, hemorrhage, and hard exudate segmentation. All methods are evaluated using the same protocol, with a frozen encoder and a trainable ConvNeXt decoder. In the full-data setting, EyeMVP achieves the highest Dice and IoU on all four datasets. The improvements are more evident on lesion segmentation tasks, where the target regions are small and often have heterogeneous appearance. On IDRID hemorrhage segmentation, EyeMVP achieves Dice/IoU of 70.86%/68.54%, compared with 61.18%/55.38% for VisionFM. On e.opthaEX hard exudate segmentation, EyeMVP achieves 82.98%/76.20%, compared with 67.50%/62.05% for VisionFM. Qualitative examples in Fig. 2 show a similar pattern: EyeMVP better preserves fine vascular structures and produces lesion masks that are more consistent with the ground-truth annotations, whereas the baseline models tend to miss small lesion regions or produce fragmented predictions.

The few-shot segmentation results show a similar trend under limited annotation. At $K=16$, EyeMVP obtains Dice/IoU of 91.71%/85.58% on REFUGE disc/cup segmentation and 90.66%/86.63% on DRIVE vessel segmentation. For lesion segmentation, EyeMVP reaches 67.92%/61.40% on IDRID hemorrhage segmentation and 67.44%/64.89% on e.opthaEX hard exudate segmentation, outperforming the competing

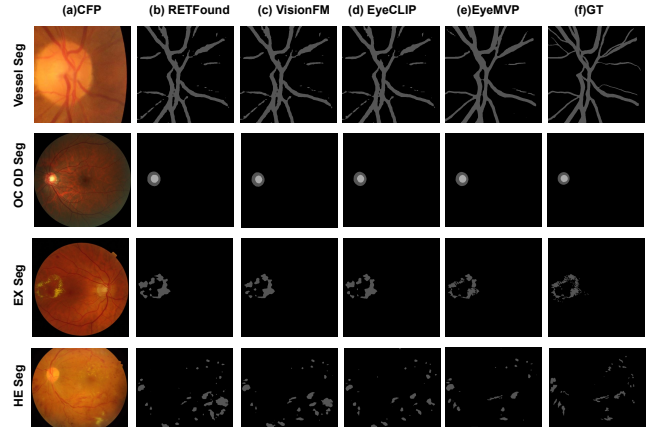


Fig. 2. Qualitative comparison of full-data segmentation results. Columns show the input CFP image, predictions from RETFound, VisionFM, EyeCLIP, EyeMVP, and the ground truth annotation. Rows correspond to retinal vessel segmentation, optic disc/cup segmentation, hard exudate segmentation, and hemorrhage segmentation. EyeMVP produces segmentation masks that are generally closer to the ground truth, particularly for small lesion regions and fine vascular structures.

foundation models under the same few-shot protocol. These results suggest that EyeMVP provides CFP representations that are useful for dense prediction tasks, including settings with limited pixel-level annotations.

F. Feature-Space Visualization

To qualitatively assess the structure and class discriminability of the learned representation space, we visualized the CFP embeddings generated by frozen pretrained encoders using t-SNE [40]. For each model, CFP images were passed through the frozen encoder, and the resulting global feature vectors were obtained directly from the encoder output without task-specific fine-tuning. These features were then projected into a two-dimensional space using t-SNE with identical hyperparameters across all methods. No label information was used during projection; ground-truth labels were used only to color the projected points. We conducted this analysis on four representative tasks: diabetic retinopathy (DR) grading and three macular conditions that are difficult to assess from CFP alone, namely age-related macular degeneration (AMD), myopic macular schisis, and macular edema. RETFound [6], EyeCLIP [8], VisionFM [7], and EyeMVP were compared under the same embedding-extraction and projection protocol.

As shown in Fig. 3, EyeMVP produces embedding spaces with more coherent class structure than the baseline foundation models. For DR grading, the embeddings of EyeMVP form more clearly organized severity-related clusters. The non-referable population is concentrated in compact regions that are better separated from higher-severity grades, whereas the baseline embeddings show greater overlap between adjacent grades. This advantage is most evident for the macular conditions. For macular edema, EyeMVP produces substantially cleaner separation between normal and edema cases, whereas RETFound and EyeCLIP show considerable intermixing between the two classes. Similar trends are observed for AMD

TABLE III

FULL-DATA AND FEW-SHOT CLASSIFICATION AUROC (%). FULL-DATA RESULTS ARE FROM LINEAR PROBING WITH THE FULL TRAINING SET. FEW-SHOT RESULTS ARE REPORTED AT $K=4, 8, 16$ LABELED SAMPLES PER CLASS. VALUES ARE MEAN $\pm$ SD. **BOLD**: BEST PERFORMANCE. AP AND BACC ARE REPORTED IN SUPPLEMENTARY TABLES S3 AND S4.

Task	Dataset	Setting	RETFound	VisionFM	EyeCLIP	EyeMVP
DR cls	Messidor-2 [30]	Full	91.21 $\pm$ 2.20	87.35 $\pm$ 3.71	80.62 $\pm$ 2.99	92.69 $\pm$ 2.02
		$K=4$	73.32 $\pm$ 7.10	69.92 $\pm$ 4.73	60.79 $\pm$ 3.08	75.52 $\pm$ 3.71
		$K=8$	77.46 $\pm$ 5.95	76.97 $\pm$ 3.46	57.23 $\pm$ 7.01	77.71 $\pm$ 4.06
		$K=16$	78.37 $\pm$ 6.35	80.34 $\pm$ 1.88	58.58 $\pm$ 5.12	83.86 $\pm$ 1.86
PDR cls	IDRID [31]	Full	82.21 $\pm$ 4.31	81.36 $\pm$ 4.02	68.91 $\pm$ 6.80	82.82 $\pm$ 3.32
		$K=4$	65.58 $\pm$ 7.98	65.57 $\pm$ 5.17	56.99 $\pm$ 8.18	75.32 $\pm$ 2.75
		$K=8$	75.30 $\pm$ 8.42	66.72 $\pm$ 7.87	57.37 $\pm$ 9.10	78.25 $\pm$ 5.93
		$K=16$	77.72 $\pm$ 5.72	68.59 $\pm$ 7.83	61.16 $\pm$ 4.59	79.75 $\pm$ 4.21
AMD cls	IChallenge-AMD [32]	Full	92.01 $\pm$ 2.74	91.76 $\pm$ 3.81	82.61 $\pm$ 7.97	94.81 $\pm$ 2.34
		$K=4$	79.32 $\pm$ 5.96	83.78 $\pm$ 3.50	70.88 $\pm$ 5.26	84.87 $\pm$ 3.31
		$K=8$	81.66 $\pm$ 8.94	85.01 $\pm$ 4.17	74.32 $\pm$ 4.83	86.22 $\pm$ 2.28
		$K=16$	87.86 $\pm$ 7.85	91.78 $\pm$ 2.30	74.45 $\pm$ 5.24	91.67 $\pm$ 1.47
Glaucoma cls	ORIGA-650 [33]	Full	72.26 $\pm$ 3.52	70.98 $\pm$ 3.38	68.84 $\pm$ 1.80	78.06 $\pm$ 2.84
		$K=4$	59.22 $\pm$ 3.65	63.89 $\pm$ 4.35	51.53 $\pm$ 3.11	66.23 $\pm$ 3.98
		$K=8$	62.68 $\pm$ 6.15	63.90 $\pm$ 5.74	53.99 $\pm$ 3.44	68.49 $\pm$ 3.89
		$K=16$	62.00 $\pm$ 4.43	60.34 $\pm$ 4.12	59.33 $\pm$ 5.60	70.13 $\pm$ 3.18
	REFUGE [34]	Full	95.87 $\pm$ 1.28	94.46 $\pm$ 2.28	85.44 $\pm$ 5.80	96.32 $\pm$ 2.15
		$K=4$	73.55 $\pm$ 5.12	75.65 $\pm$ 9.15	50.16 $\pm$ 7.21	76.70 $\pm$ 5.83
		$K=8$	81.51 $\pm$ 8.34	86.11 $\pm$ 7.40	58.78 $\pm$ 8.58	86.80 $\pm$ 4.35
		$K=16$	83.92 $\pm$ 4.80	86.83 $\pm$ 4.49	68.25 $\pm$ 2.45	87.50 $\pm$ 5.34
PM cls	PALM-2 [35]	Full	99.90 $\pm$ 0.07	99.92 $\pm$ 0.05	99.16 $\pm$ 0.60	99.86 $\pm$ 0.12
		$K=4$	99.95 $\pm$ 0.07	99.62 $\pm$ 0.12	87.34 $\pm$ 2.51	99.29 $\pm$ 0.23
		$K=8$	99.98 $\pm$ 0.02	99.79 $\pm$ 0.11	92.29 $\pm$ 2.88	99.41 $\pm$ 0.25
		$K=16$	99.97 $\pm$ 0.03	99.75 $\pm$ 0.13	94.15 $\pm$ 2.50	99.57 $\pm$ 0.19
RVO cls	Private	Full	91.55 $\pm$ 2.30	91.81 $\pm$ 2.07	79.21 $\pm$ 1.53	92.92 $\pm$ 1.78
		$K=4$	60.06 $\pm$ 6.99	64.18 $\pm$ 7.16	47.45 $\pm$ 4.11	64.09 $\pm$ 5.02
		$K=8$	63.49 $\pm$ 7.66	74.33 $\pm$ 4.43	52.90 $\pm$ 4.00	74.10 $\pm$ 4.44
		$K=16$	68.98 $\pm$ 5.48	77.89 $\pm$ 7.02	52.01 $\pm$ 3.08	78.87 $\pm$ 4.09
CSR cls	Private	Full	90.31 $\pm$ 2.92	85.70 $\pm$ 4.45	74.95 $\pm$ 3.88	91.49 $\pm$ 3.20
		$K=4$	53.56 $\pm$ 8.61	61.95 $\pm$ 6.26	48.01 $\pm$ 8.51	64.49 $\pm$ 4.23
		$K=8$	57.02 $\pm$ 10.30	67.08 $\pm$ 8.54	48.98 $\pm$ 4.64	67.91 $\pm$ 5.92
		$K=16$	64.90 $\pm$ 6.59	71.21 $\pm$ 3.35	52.96 $\pm$ 6.93	73.45 $\pm$ 3.78
Multi-disease cls	ODIR5K [36]	Full	77.28 $\pm$ 0.68	75.13 $\pm$ 0.77	69.21 $\pm$ 0.59	77.37 $\pm$ 0.99
		$K=4$	60.25 $\pm$ 1.71	64.12 $\pm$ 2.60	53.16 $\pm$ 1.17	65.37 $\pm$ 2.90
		$K=8$	62.76 $\pm$ 1.06	66.70 $\pm$ 1.04	56.15 $\pm$ 2.16	67.06 $\pm$ 2.04
		$K=16$	66.56 $\pm$ 1.89	69.31 $\pm$ 1.49	58.54 $\pm$ 3.69	72.14 $\pm$ 1.62
ME cls	Private	Full	90.89 $\pm$ 1.77	88.78 $\pm$ 2.65	73.95 $\pm$ 2.59	92.26 $\pm$ 2.88
		$K=4$	62.87 $\pm$ 10.22	63.13 $\pm$ 6.66	47.01 $\pm$ 12.29	68.57 $\pm$ 6.08
		$K=8$	68.64 $\pm$ 8.48	68.24 $\pm$ 5.20	48.90 $\pm$ 15.44	71.35 $\pm$ 4.85
		$K=16$	74.98 $\pm$ 4.55	75.02 $\pm$ 2.76	52.93 $\pm$ 12.11	81.71 $\pm$ 2.14
MS cls	Private	Full	76.29 $\pm$ 2.91	82.00 $\pm$ 3.05	75.10 $\pm$ 7.79	86.67 $\pm$ 2.10
		$K=4$	64.19 $\pm$ 8.70	61.74 $\pm$ 16.26	48.45 $\pm$ 8.21	68.57 $\pm$ 5.14
		$K=8$	72.88 $\pm$ 2.28	71.81 $\pm$ 7.73	57.24 $\pm$ 12.53	74.10 $\pm$ 3.67
		$K=16$	79.02 $\pm$ 4.04	71.76 $\pm$ 5.32	60.43 $\pm$ 9.26	83.88 $\pm$ 3.11

DR: diabetic retinopathy; PDR: proliferative diabetic retinopathy; AMD: age-related macular degeneration; PM: pathological myopia; RVO: retinal vein occlusion; CSR: central serous retinopathy; ME: macular edema; MS: myopic macular schisis.

and myopic macular schisis, where the embeddings of EyeMVP form more contiguous disease regions with reduced overlap with the normal population.

These qualitative observations are consistent with the quantitative linear-probing results in Table III. The macular conditions on which EyeMVP shows clear AUROC gains, including myopic macular schisis and macular edema, also show the most visibly improved class separation in the embedding space. Because the projection uses only the output of the frozen encoder without supervised adaptation, the improved cluster structure suggests that cross-modal CFP–OCT pretraining enables CFP representations to encode disease-relevant structural information in a more linearly separable form. This finding is consistent with the improved label efficiency observed in the few-shot experiments. We note that t-SNE is a nonlinear projection method intended for qualitative

inspection, and absolute inter-cluster distances should not be interpreted quantitatively.

G. Reader Study

We conducted an exploratory reader study on an independent test set of 150 cases, including 50 myopic macular schisis, 50 macular edema, and 50 control eyes. The test set was not used for pretraining, model selection, or downstream fine-tuning. Nine ophthalmologists independently evaluated the CFP images and were stratified by clinical experience into junior (<5 years), intermediate (5–10 years), and senior (>10 years) groups, with three readers in each group. Balanced accuracy (BACC) was computed for each reader and averaged within each experience group. EyeMVP was evaluated on the same test set using CFP images only.

TABLE IV

FULL-DATA AND FEW-SHOT SEGMENTATION PERFORMANCE. FULL-DATA RESULTS ARE OBTAINED WITH A FROZEN ENCODER AND A TRAINABLE CONVNEXT DECODER. FEW-SHOT RESULTS ARE REPORTED AT $K=4, 8, 16$ LABELED SAMPLES. DICE AND IOU ARE REPORTED IN PERCENTAGE. **BOLD**: BEST PERFORMANCE. QUALITATIVE FEW-SHOT EXAMPLES ARE SHOWN IN SUPPLEMENTARY FIG. S1.

Dataset (Task)	Setting	Metric	RETFound	VisionFM	EyeCLIP	EyeMVP
REFUGE [34] (Disc/Cup)	Full	Dice	88.44 $\pm$ 0.50	91.15 $\pm$ 0.42	87.13 $\pm$ 0.41	94.48 $\pm$ 0.36
		IoU	80.29 $\pm$ 0.62	84.37 $\pm$ 0.49	79.34 $\pm$ 0.49	85.82 $\pm$ 0.36
	$K=4$	Dice	71.69 $\pm$ 4.12	74.36 $\pm$ 3.85	77.56 $\pm$ 3.24	84.33 $\pm$ 2.95
		IoU	62.80 $\pm$ 4.45	67.66 $\pm$ 4.10	60.27 $\pm$ 3.44	78.76 $\pm$ 3.20
	$K=8$	Dice	79.59 $\pm$ 2.21	81.45 $\pm$ 1.95	78.37 $\pm$ 1.88	89.51 $\pm$ 1.15
		IoU	68.77 $\pm$ 2.45	71.02 $\pm$ 2.15	64.45 $\pm$ 2.05	82.23 $\pm$ 1.40
	$K=16$	Dice	86.32 $\pm$ 1.45	90.31 $\pm$ 1.12	83.55 $\pm$ 1.52	91.71 $\pm$ 0.85
		IoU	77.24 $\pm$ 1.68	83.09 $\pm$ 1.34	73.92 $\pm$ 1.25	85.58 $\pm$ 0.95
DRIVE [37] (Vessels)	Full	Dice	85.10 $\pm$ 0.45	87.66 $\pm$ 0.38	84.91 $\pm$ 0.40	91.90 $\pm$ 0.25
		IoU	76.80 $\pm$ 0.52	79.54 $\pm$ 0.45	76.55 $\pm$ 0.46	87.25 $\pm$ 0.32
	$K=4$	Dice	78.90 $\pm$ 3.15	84.49 $\pm$ 2.82	79.48 $\pm$ 2.95	88.65 $\pm$ 2.25
		IoU	69.35 $\pm$ 3.40	75.35 $\pm$ 3.05	68.85 $\pm$ 3.15	83.74 $\pm$ 2.45
	$K=8$	Dice	79.88 $\pm$ 2.65	84.71 $\pm$ 2.42	80.07 $\pm$ 2.50	89.12 $\pm$ 1.95
		IoU	70.42 $\pm$ 2.88	75.71 $\pm$ 2.65	70.87 $\pm$ 2.72	83.96 $\pm$ 2.12
	$K=16$	Dice	81.05 $\pm$ 1.10	84.73 $\pm$ 0.95	80.62 $\pm$ 1.02	90.66 $\pm$ 0.65
		IoU	71.98 $\pm$ 1.25	75.75 $\pm$ 1.10	71.39 $\pm$ 1.18	86.63 $\pm$ 0.78
IDRID [31] (HE)	Full	Dice	59.50 $\pm$ 1.25	61.18 $\pm$ 1.15	56.85 $\pm$ 1.30	70.86 $\pm$ 0.95
		IoU	53.55 $\pm$ 1.35	55.38 $\pm$ 1.22	52.72 $\pm$ 1.42	68.54 $\pm$ 1.10
	$K=4$	Dice	49.90 $\pm$ 2.15	50.80 $\pm$ 2.45	49.50 $\pm$ 1.82	58.28 $\pm$ 1.45
		IoU	45.30 $\pm$ 2.41	46.12 $\pm$ 2.55	47.75 $\pm$ 1.65	52.85 $\pm$ 1.72
	$K=8$	Dice	52.48 $\pm$ 1.85	53.78 $\pm$ 1.64	51.20 $\pm$ 1.75	63.09 $\pm$ 1.25
		IoU	47.62 $\pm$ 2.05	48.80 $\pm$ 1.85	48.10 $\pm$ 1.54	57.12 $\pm$ 1.45
	$K=16$	Dice	55.96 $\pm$ 1.35	56.85 $\pm$ 1.20	53.75 $\pm$ 1.28	67.92 $\pm$ 0.85
		IoU	50.75 $\pm$ 1.52	51.55 $\pm$ 1.42	50.15 $\pm$ 1.48	61.40 $\pm$ 1.05
e_opthaEX [38] (EX)	Full	Dice	63.90 $\pm$ 0.93	67.50 $\pm$ 0.92	60.10 $\pm$ 0.39	82.98 $\pm$ 0.75
		IoU	56.79 $\pm$ 0.98	62.05 $\pm$ 0.76	58.68 $\pm$ 0.41	76.20 $\pm$ 0.88
	$K=4$	Dice	51.26 $\pm$ 2.52	53.83 $\pm$ 2.15	49.18 $\pm$ 2.65	63.67 $\pm$ 1.25
		IoU	46.55 $\pm$ 2.80	48.34 $\pm$ 2.40	48.38 $\pm$ 1.92	58.59 $\pm$ 1.55
	$K=8$	Dice	54.75 $\pm$ 1.85	58.34 $\pm$ 1.65	53.21 $\pm$ 2.05	64.38 $\pm$ 0.85
		IoU	49.68 $\pm$ 2.10	54.38 $\pm$ 2.92	51.08 $\pm$ 1.30	59.54 $\pm$ 1.15
	$K=16$	Dice	59.18 $\pm$ 1.20	62.65 $\pm$ 1.10	56.85 $\pm$ 1.55	67.44 $\pm$ 0.67
		IoU	53.65 $\pm$ 1.45	56.55 $\pm$ 1.30	54.86 $\pm$ 1.78	64.89 $\pm$ 0.45

HE: hemorrhage; EX: hard exudate.

As shown in Table V, EyeMVP (86.37%) exceeded the junior (74.08%) and intermediate (82.10%) groups but did not reach the performance of senior ophthalmologists (90.12%) on macular edema. For myopic macular schisis, EyeMVP achieved a BAcc of 61.46%, compared with 44.04%, 50.00%, and 55.36% for the junior, intermediate, and senior groups, respectively. These exploratory results suggest that EyeMVP may provide complementary CFP-based information for macular conditions that are difficult to assess from fundus photographs alone. Given the limited size of the reader study, these findings should be interpreted as preliminary and require validation in larger multi-reader cohorts.

H. Ablation Studies

Table VI reports component ablations on three tasks where retinal or macular structure is expected to be important: ORIGA-650 glaucoma, macular edema, and myopic macular schisis. Each ablation removes one pretraining component while using the same CFP encoder architecture and otherwise matched pretraining and downstream linear probing conditions. We report representative ablations in the main text; complete results across all tasks are provided in Supplementary Table S5.

TABLE V

READER STUDY ON BALANCED ACCURACY (BACC, %) FOR MYOPIC MACULAR SCHISIS (MS) AND MACULAR EDEMA (ME). NINE OPHTHALMOLOGISTS WERE STRATIFIED BY EXPERIENCE LEVEL ($n=3$ PER GROUP) AND COMPARED WITH EYEMVP ON THE SAME 150-CASE INDEPENDENT TEST SET.

Reader group	MS	ME
Junior (<5 years)	44.04	74.08
Intermediate (5–10 years)	50.00	82.10
Senior (>10 years)	55.36	90.12
EyeMVP	61.46	86.37

Removing the OCT modality from pretraining leads to the largest performance decrease across the three tasks, with AUROC drops of 5.53, 11.60, and 12.57 percentage points on ORIGA-650, macular edema, and myopic macular schisis, respectively. This result supports the role of paired CFP–OCT data in providing useful cross-modal supervision for CFP representation learning.

The Stage-I-only variant retains the shared encoder and the three intra-modal reconstruction decoders, but removes the source-constrained CFP-to-OCT and OCT-to-CFP reconstruction objectives introduced in Stage II. This setting corresponds to a shared-encoder multimodal masked autoencoding base-

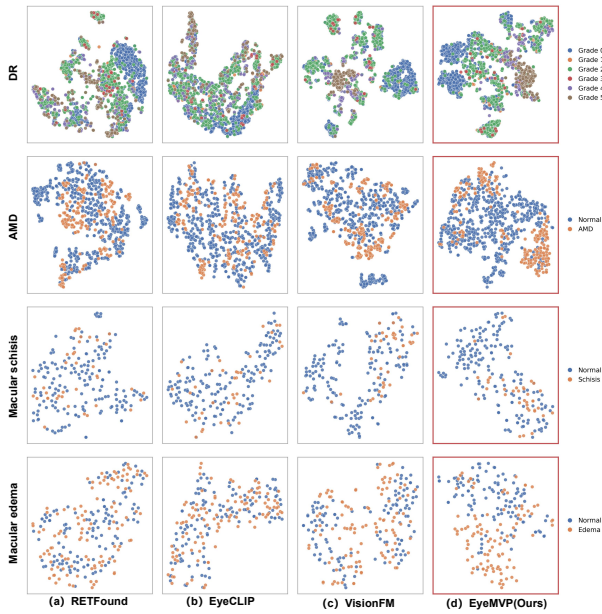


Fig. 3. t-SNE visualization of frozen CFP encoder embeddings for RETFound, EyeCLIP, VisionFM, and EyeMVP (Ours). Each point corresponds to the encoder output embedding of one CFP image projected to two dimensions; colors denote ground-truth class labels and are not used during projection. Rows correspond to four tasks: DR (six severity grades), AMD (normal vs. AMD), myopic macular schisis (normal vs. schisis), and macular edema (normal vs. edema). Compared with the baseline foundation models, EyeMVP produces more class-coherent clusters with reduced overlap between disease and normal populations, with the clearest improvements on the macular conditions.

line, where each modality is reconstructed primarily from its own visible tokens. Compared with the full two-stage model, Stage-I-only pretraining reduces AUROC by 4.33, 7.90, and 6.12 percentage points on ORIGA-650, macular edema, and myopic macular schisis, respectively. These results indicate that explicit cross-modal reconstruction provides additional supervision beyond shared-encoder intra-modal reconstruction.

To further examine this component, we visualize CFP-to-OCT reconstruction under a source-constrained setting in Fig. 4, where all OCT tokens are masked and only partial CFP tokens are provided. Compared with the Stage-I-only variant, EyeMVP produces reconstructions with clearer OCT-like retinal contours and layer-related contrast. These visualizations illustrate the pretraining objective and are not intended to represent diagnostic OCT synthesis.

Removing the CFP-Seg branch decreases AUROC by 4.91 percentage points on ORIGA-650 and 4.03 percentage points on myopic macular schisis. This suggests that CFP-derived anatomical masks provide useful spatial guidance during pretraining by supplying coarse fundus-layout context for cross-modal reconstruction. Overall, the ablation results show that the paired OCT modality, Stage-II cross-modal reconstruction, and CFP-derived structural masks each contribute to the final EyeMVP performance.

V. DISCUSSION

This study presents EyeMVP, a cross-modal retinal foundation model that uses paired CFP–OCT data to improve

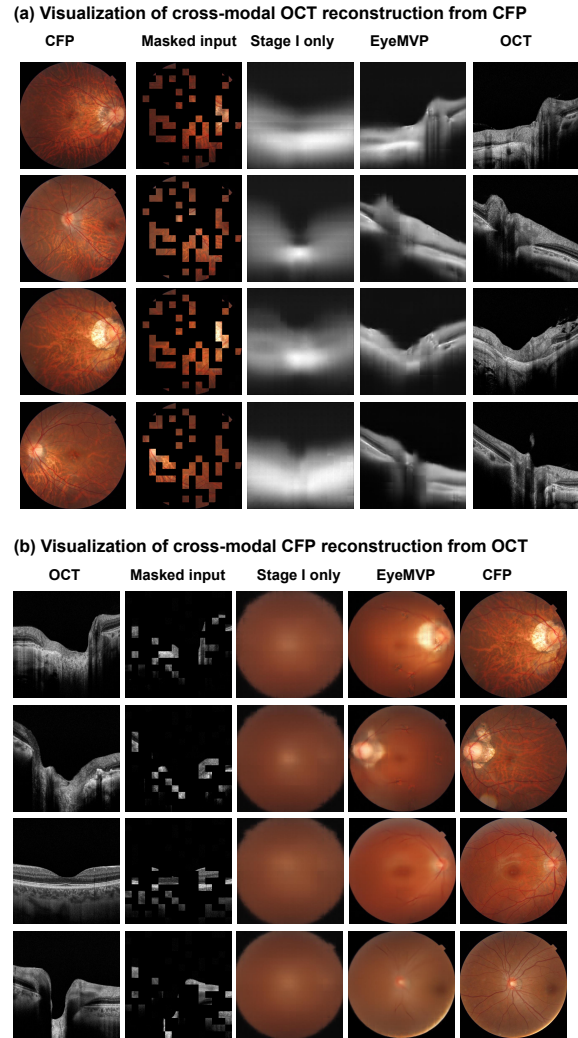


Fig. 4. Qualitative visualization of bidirectional cross-modal reconstruction during pretraining. (a) OCT reconstruction from partially visible CFP tokens. Columns show the CFP image, masked CFP input, Stage-I-only reconstruction, EyeMVP reconstruction with source-constrained cross-modal decoding, and the paired OCT target. (b) CFP reconstruction from partially visible OCT tokens, shown in the reverse direction with the paired fundus image as the target reference. Compared with the Stage-I-only variant, EyeMVP recovers sharper modality-specific structures, including OCT-like retinal contours and more fundus-like vascular and disc appearance. These visualizations illustrate the cross-modal pretraining objective and are not intended to represent diagnostic OCT or CFP synthesis.

CFP-based representation learning. Across classification, segmentation, few-shot learning, feature-space visualization, and reader-study evaluations, EyeMVP shows consistent gains over representative retinal foundation models. The improvements are most evident on tasks in which retinal or macular structure is clinically relevant but not directly resolved in CFP, such as glaucoma, macular edema, and myopic macular schisis. These findings support the value of using OCT as a structural supervisory signal during CFP pretraining, while retaining the practical advantage that only CFP images are required at inference.

The design of EyeMVP reflects several practical challenges in CFP–OCT pretraining. CFP and OCT are acquired in

TABLE VI

COMPONENT ABLATION AUROC (%) ON STRUCTURALLY RELEVANT CLASSIFICATION TASKS. VALUES IN PARENTHESES DENOTE ABSOLUTE AUROC DROP (Δ) RELATIVE TO THE FULL MODEL. MEAN VALUES REPORTED; SD $\leq 2.91\%$ ACROSS ALL ENTRIES.

Configuration	ORIGA-650	ME	MS
w/o OCT modality	72.53 (−5.53)	80.66 (−11.60)	74.10 (−12.57)
w/o Stage-II cross recon.	73.73 (−4.33)	84.36 (−7.90)	80.55 (−6.12)
w/o CFP-Seg	73.15 (−4.91)	85.23 (−7.03)	82.64 (−4.03)
EyeMVP (full)	78.06	92.26	86.67

different imaging geometries, and their image tokens do not share a common spatial grid. A purely intra-modal masked autoencoding objective may therefore learn modality-specific reconstruction shortcuts without sufficiently encouraging cross-modal interaction. EyeMVP addresses this issue using source-constrained cross-modal decoding, which requires one modality to predict the other from source-modality representations. The CFP-Seg branch provides coarse anatomical guidance for the fundus image, and the Dirichlet masking strategy exposes the model to variable modality availability during pretraining. Ablation results and reconstruction visualizations suggest that these components contribute complementary benefits. We use the two-stage schedule as an implementation choice to stabilize optimization rather than as an independent methodological contribution.

EyeMVP should not be interpreted as a replacement for OCT. The model does not measure retinal depth or synthesize diagnostic OCT images; instead, it learns CFP representations that are informed by statistical associations with paired OCT data. This distinction is important for clinical deployment. In settings where OCT is unavailable, such representations may help improve CFP-based screening and triage, but positive or uncertain findings should still be confirmed with standard clinical examination and OCT when indicated. The exploratory reader study further suggests that EyeMVP may provide complementary CFP-based information for macular conditions that are difficult to assess from CFP alone, but larger multi-reader and prospective studies are required before drawing clinical conclusions.

Despite its strong performance, we acknowledge the methodological limits of the approach. EyeMVP learns statistical CFP–OCT structural correspondence during cross-modal pretraining rather than a registered or physically grounded mapping, relying on fine-grained texture and vascular patterns as learned surrogates for retinal structure. Because the CFP–OCT pairs are linked at the eye-visit rather than the pixel level, it captures population-level correspondence rather than exact spatial alignment, a trade-off that enables large-scale pretraining while bounding spatial precision. As these learned correlations are bounded by the pretraining distribution, performance may degrade when the underlying cues are disrupted, such as under media opacities (e.g., dense cataracts or vitreous hemorrhage) or in rare phenotypes and post-surgical eyes that violate normal CFP–OCT relationships. Future deployments should incorporate uncertainty quantification to flag such out-of-distribution cases for clinical review.

VI. CONCLUSION

We presented EyeMVP, a cross-modal retinal foundation model that exploits paired CFP–OCT data during pretraining to enrich CFP representations while requiring only CFP images at inference. Its core novelty is a source-constrained cross-modal reconstruction scheme that forces CFP features to predict OCT structure without access to target-modality tokens, complemented by an auxiliary CFP-Seg branch that injects anatomical-layout priors. Across classification, dense segmentation, and few-shot settings, EyeMVP is competitive with or better than representative retinal foundation models, with the clearest gains on glaucoma and macular conditions, where structural cues are clinically important but poorly resolved in CFP. By transferring OCT-derived structural information into a CFP-only model, EyeMVP offers a practical path to structure-aware fundus analysis where OCT is unavailable.

DECLARATIONS

A. Data and Code availability

The pre-training data was collected in-house and is not publicly available due to privacy restrictions. Downstream evaluation was conducted on 11 public datasets and 4 private in-house datasets. The public datasets can be accessed via the following links: Messidor-2 [30] (<https://www.adcis.net/en/third-party/messidor2/>), IDRID [31] (<http://idrid.grand-challenge.org/grading>), IChallenge-AMD [32] (<https://amd.grand-challenge.org/>), ORIGA-650 [33] (<https://doi.org/10.57702/hoijctum>), REFUGE [34] (<https://refuge.grand-challenge.org/Home2020/grading>), PALM [35] (<https://palm.grand-challenge.org/>), ODIR5K [36] (<https://odir2019.grand-challenge.org/introduction/>), DRIVE [37] (<https://drive.grand-challenge.org/>), REFUGE [34] (<https://refuge.grand-challenge.org/Home2020/segmentation>), e.opthaEX [38] (<https://www.adcis.net/en/third-party/e-optha/>), and IDRID [31] (<http://idrid.grand-challenge.org/segmentation>). While the 4 in-house evaluation datasets cannot be fully shared due to patient privacy regulations. The dataset, code, and models will be publicly available at <https://github.com/ML-AILab/EyeMVP>.

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